THE PATH TOWARDS A TECHNOLOGICAL INTERVENTION: EVALUATING VERIFICATION AND VALIDATION METHODS*

Introduction to Artificial Intelligence
30 September 2024

Dr. Sara Lusini
s.lusini@vu.nl

**These slides are based on a previous presentation by Dr. M. Klein.*

CONTENT

- **PART 1**
  - Learning Objectives
  - The path towards a Technological Intervention
  - Verification vs Validation
  - Exercise 1 (about 10 minutes preparation + 10 minutes discussion)

  BREAK (10 minutes)

- **PART 2**
  - External Validation
  - Experimental vs Control group
  - Evaluating intervention effects and statistical relevance

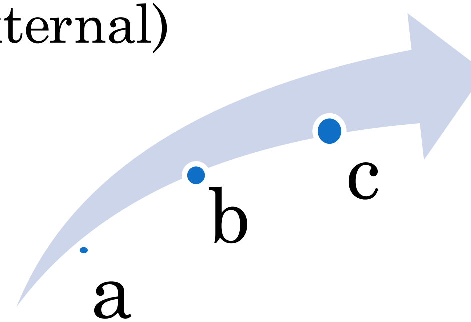
- **PART 3**
  - Setting up a (schematic) Research Protocol
  - External Validation of Research Protocol:
    - Validation (measure impact of intervention)
    - User evaluation
  - Exercise 2 (about 10 minutes preparation + 10 minutes discussion)

PART 1

Learning objectives of this lecture

Upon completion of this lecture, students will be able to:

1. Use the scientific terminology relative to Verification & Validation Methods and their evaluation.
2. Design a basic Research Protocol for the (external) validation of a Technological Intervention.



How will we get there?

Here is the process step-by-step:

- The bigger picture: provide an overview of the Technological Intervention process.
- Discuss and understand the difference between Validation and Verification, and why they are both needed.
- Discuss examples of Research Protocols for the (external) validation of a Technological Intervention.

Technological intervention…what's that?

- Think about a lifestyle or medical app **supporting health** that you used and did not work well for you (e.g. related to exercise, nutrition, sleep, meditation, blood sugar measurement etc.). Think of **why it didn't work**.

EXAMPLE: AN APP THAT HELPS PEOPLE TO STOP SMOKING AND PROVIDES USERS WITH A PERSONALIZED PLAN AND ADVICE

INTERVENTION SYSTEM

Designing the System

Define system requirements and target groups.

Implementing the System

Design System Architecture and Inference Engine, Code.

Testing the System

Evaluate **Verification** and fine-tune parameters

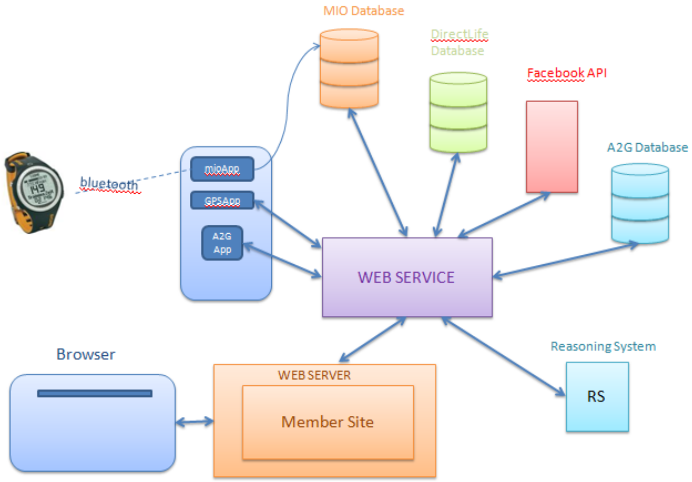
Using the System

Validation + user evaluations

THE PATH TOWARDS A TECHNOLOGICAL INTERVENTION

- **System design**
  - System Requirements
  - Target groups/test subjects

- **System implementation**
  - System Architecture
  - Inference Engine
  - Programming



- **System Testing (verification)**
  - Do the components work?
  - Does the system give the expected output based on a given input?

- **System Validation**
  - Does the system help its users achieve their goal?
  - How is the overall user experience?

VERIFICATION & VALIDATION

Verification is the process of checking whether the software system meets the specified requirements of the users ☞ Testing the system's *Correctness:* Is the system doing what it is supposed to be doing? Can it correctly keep track of how much the users are smoking? Is it able to provide users with personalized plans targeting individual users' need?

Validation is the process of checking whether the software system meets the actual requirements of the users ☞ Testing the system's *Effectiveness* Does the system actually work? Does it help its users smoke less or quit smoking?

Internal vs external validation

Internal validation

External validation

Building the System

Using the System

Verification
Components/models
Simulations
Parameter tuning

Validation
Measure the impact/effect of the intervention/ (user) evaluations

Exercise 1:
set up intervention

First, think of a new **Lifestyle and/or Medical Intervention App Supporting Health Care**.

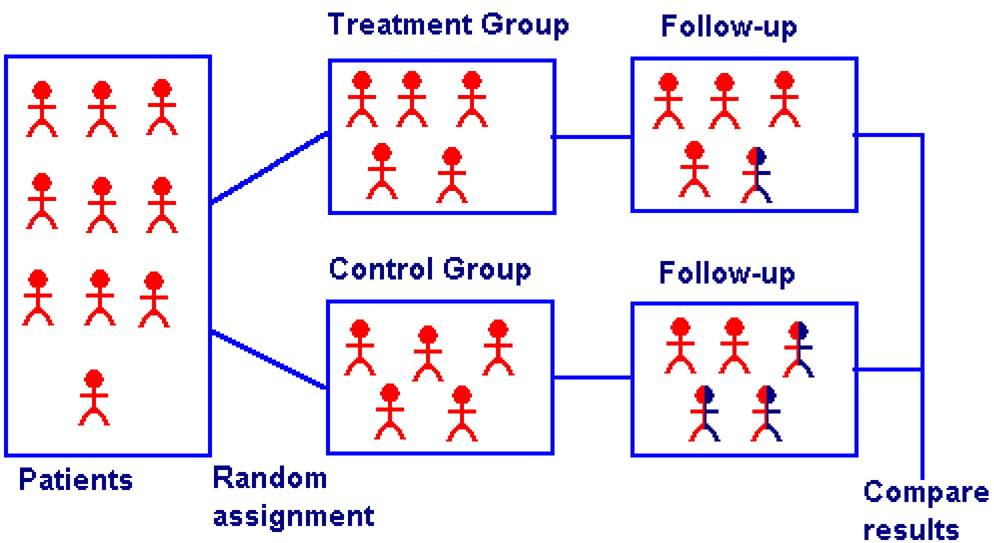
Then, discuss **how you could actually test** if the app works or not (i.e. whether it achieves the intended goals).

After 10 minutes of discussion, I will ask a couple of groups to share their ideas with the rest of the class.

PART 2

EXTERNAL VALIDATION (OR EFFECTIVENESS)

- External validation is best achieved by means of a Randomized Controlled Trial:



Experimental and control group

Control Group

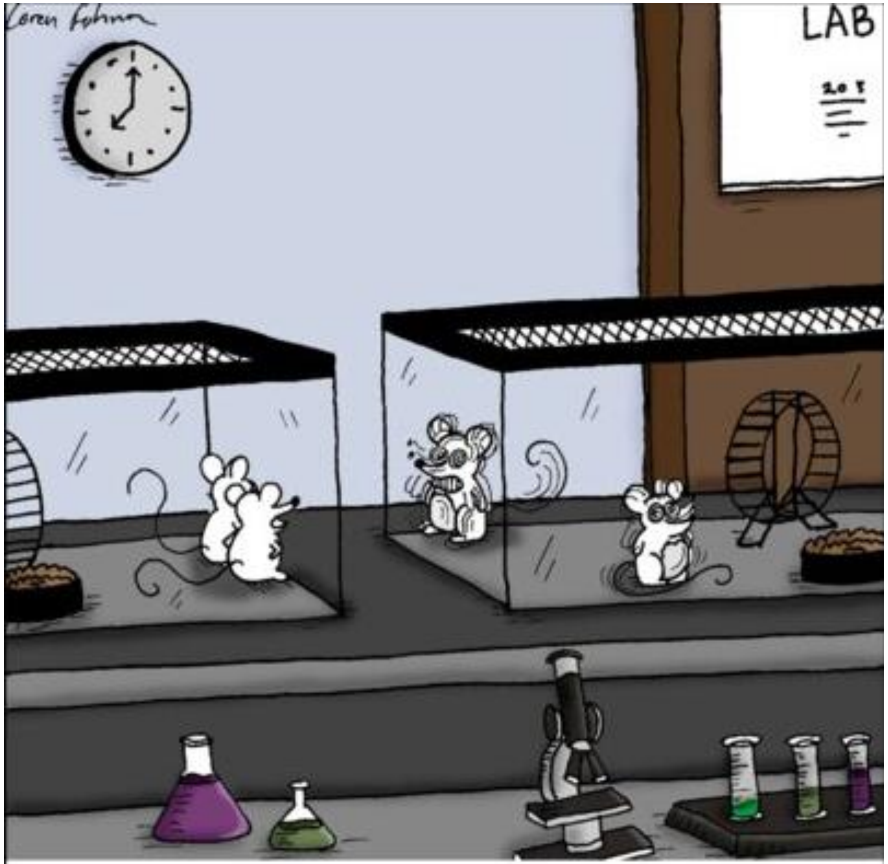
- the control group (or control condition) refers to the group of participants who are not exposed to the technological intervention.

Experimental Group

- the experimental group (or experimental condition) refers to the group of participants who are exposed to the technological intervention.

THE CONTROL GROUP

- N.B.= During **Clinical trials**, not getting any treatment is often not an option for the participants.



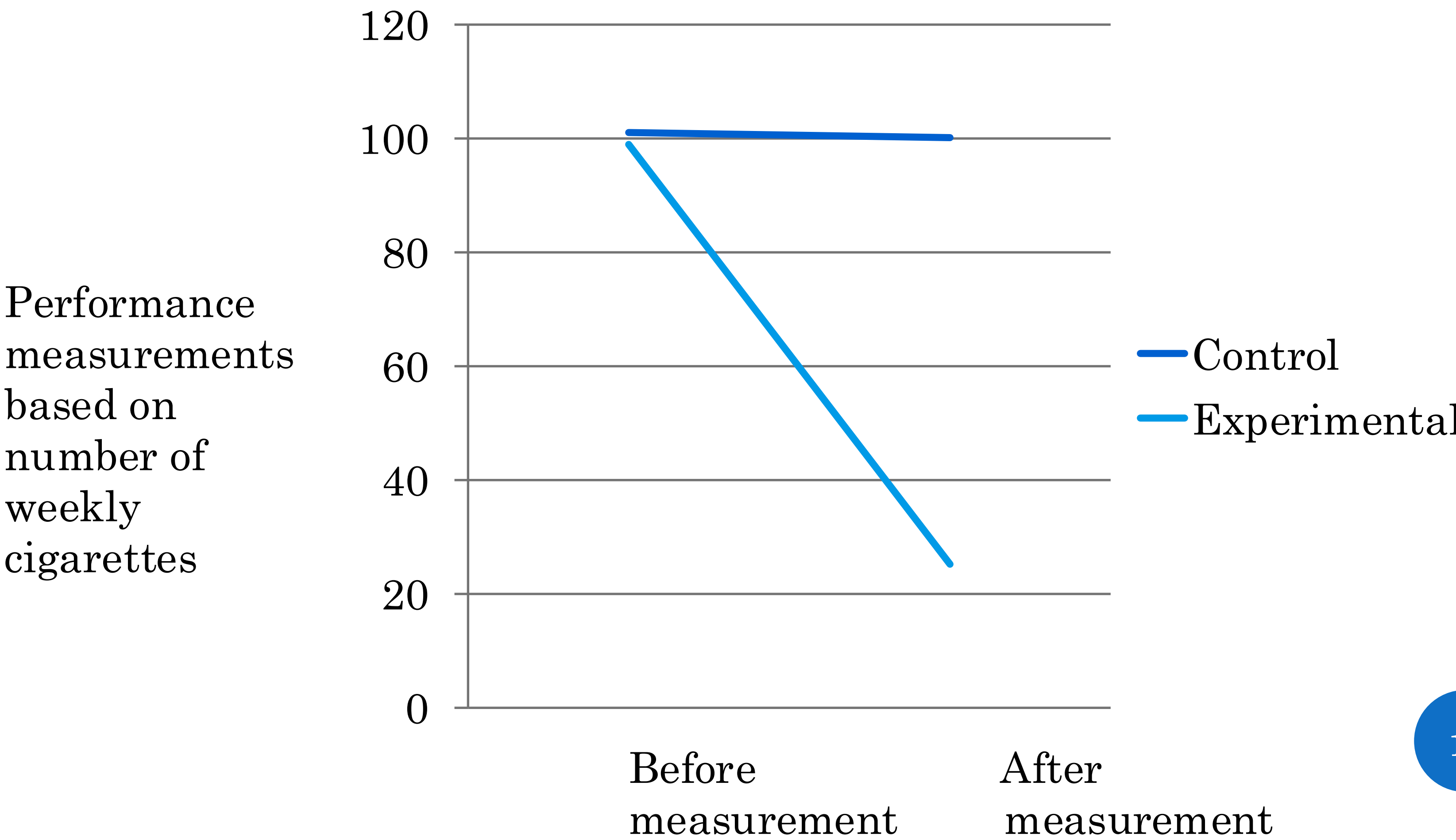
"Well, I guess we're the control group."

Why do you think this is the case?

How could you solve this issue?

EVALUATING INTERVENTION EFFECTS

If this was the outcome of your external validation process, how could you tell if the difference is significant?



IS THE DIFFERENCE SIGNIFICANT OR DUE TO CHANCE?

- You need to establish this through a statistical analysis, for instance:
  - T-tests
  - repeated measures design
  - ANOVA (Analysis of Variance)
  - ANCOVA (Analysis of Covariance)

- The primary output of statistical tests is the p-value (probability value). It indicates the **probability** of observing the difference if no difference exists.
- These tests are widely used in a.o. Social Sciences & Economics
- The easiest type of statistical test is a t-test comparing the performances of two groups of users (one measurement each).

Subjective or objective measurement?

- NB:= You can track the number of cigarettes smoked by participants through self-report, subjective measurements (e.g. questionnaires), BUT at the same time you should use more objective ways to track these data as well.

How can we measure smoking behavior in an objective way?

- A sensor that keeps track of every time a cigarette package is opened? Or a sensor that detects cigarette smoke in the proximity of the participant's faces?
- Measuring people's behavior can be complicated. If possible, multiple sensors should be used, along with self-report measurements.

PART 3
RESEARCH PROTOCOL

You need to become familiar with the following terms, which you will use to write up your Research Protocol:

- **Research Question (**is the objective of a study or a problem to be solved through research)
- **Hypothesis** (is a proposed explanation for a phenomenon)
- **Experimental and Control group** (respectively, the group of participants who are exposed to the technological intervention and the group of participants who are not exposed to the technological intervention)
- **Independent Variable** (the variable that is changed or controlled in a scientific experiment to test the effects on the dependent variable.)
- **Dependent variable (**the variable being tested and measured in a scientific experiment.)
- **Subjects/participants** (the people taking part in the experiment/intervention)
- **Possible Confounding variables (**extra independent variables that are having a hidden effect on your dependent variables)

Example : (schematic) research protocol

- **Research Question**: Is it possible to reduce smoking behavior by using a smart app that gives tailored advice to its users?
- **Hypothesis**: A smart app reduces smoking behavior more than an app that gives only information to its users, and little to no tailored advice.
- **Control Condition and Experimental Condition:** Half of the test subjects are given an app that provides only information to its users, and little to no tailored advice, while the remaining half is given a smart app that gives tailored advice to its users.
- ***Independent Variable***: *Intervention (with or without the smart app)*
- ***Dependent variable***: *smoking behavior (e.g. number of weekly cigarettes)*
- **Test Subjects**: E.g. Male, Dutch High educated, have been smoking for more than 10 years.
- **Possible Confounding Variables**: E.g. the interface is not user-friendly, and therefore the participants stop using the app.

How can you test a research question?

'Example: Does the use of a running app lead to more exercise?'

Here are some examples:

A. By letting 100 test subjects use the running app and then asking them whether it worked ot not.
B. By letting 100 test subjects use the running app and tracking their physical activity through a sensor while they are using the app.
C. By letting 100 test subjects use the running app, and at the same time letting 100 tests subjects use a puzzle app. Then, the physical activity of both groups can be tracked, and if there is a difference, it can be investigated whether it is statistically significant or not.
D. By letting 100 test subjects use the running app, and at the same time sending 100 tests subjects information materials through the mail. Then, the physical activity of both groups can be tracked, and if there is a difference, it can be investigated whether it is statistically significant or not.

User experience evaluation

Your Research protocol needs to be externally validated through **User Experience Evaluation**.

- User Experience (Usability): Do you think it works? Is it easy to use? Will you keep using it? Do you like it? Is it cost-effective and feasible?

- Questionnaires (often based on a **likert-scale**) are used to evaluate user experience . Examples include:
  - Acceptability questionnaire
  - Usability questionnaire
  - Feasibility questionnaire

Acceptability, usability and feasibility

- **Acceptability**: Acceptability refers to the factors that affect the users' willingness to use the application.

- **Usability**: Usability refers to the design factors that affect the user experience of operating the application's device and navigating the application for its intended purpose.

- **Feasibility**: Feasibility refers to whether the intervention can actually be carried out or not.

Example of usability questionnaire for 'sitcoach' system

Table 1 Names, descriptions, and examples of the four subscales of the Attrakdiff2 questionnaire

Quality	Description	Examples
Pragmatic quality	Does a product do what it's supposed to do?	Technical, human
	Is it easy to understand and use the product?	Complicated, simple
	Do users find it easy to interact with SitCoach, and does it perform according to their expectancies?	
Hedonic stimulation	Does the product help the user in developing new skills or gaining knowledge?	Typical, original
	Does SitCoach make people aware of their sedentary behavior and help them to take more frequent breaks from sitting?	Easy, challenging
Hedonic identification	How is the product used in a social context?	Isolating, integrating
	How does the product contribute to one's identity?	Cheap, valuable
	What does one communicate to others by using SitCoach	
Attractiveness	Global appeal of the product.	Ugly, beautiful
	Do users find SitCoach as a whole appealing or attractive?	Bad, Good

Exercise 2:
SET UP RESEARCH PROTOCOL

Exercise: get back to the technological intervention you had thought of for the first exercise. Use the categories below to write up **a schematic research protocol to test the (external) validity of your technological intervention:**

- Research questions
- Hypothesis
- Conditions: Control & Experimental
- Independent variable
- Dependent variable
- Test subjects
- Possible confounding variables

Do not forget to provide information about the questionnaire that will be used to **evaluate your system** (acceptibility/feasibility/cost-effectiveness, etc.).Which categories are you going to investigate? What do these questions lists measure? Provide examples.

After **10 minutes**, I will pick a couple of groups and ask them to present their research protocol to the class.


The End

References

- Hand DJ, Khan S. Validating and Verifying AI Systems. Patterns (N Y). 2020 Jun 12;1(3):100037. doi: 10.1016/j.patter.2020.100037. PMID: 33205105; PMCID: PMC7660449.
- Myllyaho, L., Raatikainen, M., Männistö, T., Mikkonen, T., & Nurminen, J. K. (2021). Systematic literature review of validation methods for AI systems. *Journal of Systems and Software*, *181*, 111050. https://doi.org/10.1016/j.jss.2021.111050
- Preece, A. (2001). Evaluating Verification and Validation Methods in Knowledge Engineering. In: Roy, R. (eds) Industrial Knowledge Management. Springer, London. https://doi.org/10.1007/978-1-4471-0351-6_6
- van Dantzig, S., Geleijnse, G. & van Halteren, A.T. Toward a persuasive mobile application to reduce sedentary behavior. *Pers Ubiquit Comput* **17**, 1237–1246 (2013). https://doi.org/10.1007/s00779-012-0588-0